Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	Low-Power, High-Resolution Multimodal Biosensing System with On-Device AI for Wearable Digital Healthcare			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio (v), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI (v), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Electrical / Electronics		Bio / Medical	
	Medium Category	Small Category	Medium Category	Small Category
	Semiconductor device and systems	SoC	Therapeutic devices & Diagnostic devices	Bio signal measurement / Diagnostic devices
Concept of Proposed Technology	The proposed technology is a **low-power, high-resolution multimodal biosensing system** designed to enable accurate and continuous monitoring of vital physiological signals in wearable digital healthcare devices.[1], [2], [3] Current wearable healthcare solutions face critical limitations due to **insufficient sensing accuracy and short battery life**, which reduce clinical reliability and user acceptance. These challenges cannot be resolved by biosensors or signal processing alone. Instead, they require an integrated approach combining **high-precision biosensors, ultra-low-power signal processing chips, and intelligent on-device data processing**. The proposed system integrates: • Multiple biosensors capable of monitoring diverse vital signals • A single-die, multimodal bio-signal processing chip with ultra-low current detection resolution and high noise immunity [4]			

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	• On-device AI algorithms that enable intelligent preprocessing and decision-making at the data acquisition stage By embedding AI directly at the sensing level, the system minimizes unnecessary data transmission, improves signal quality, and extends battery life, thereby addressing key limitations of existing wearable healthcare technologies.
Rationale for Joint R&D Project	Korean semiconductor companies are competitive in processes, components, and design. They are particularly world-leading in memory semiconductors, and they also boast significant competitiveness in various non-memory semiconductor sectors. The bio and digital healthcare sectors are poised for innovative advancements when AI and semiconductors are combined. The United States boasts unparalleled competitiveness in bio, medical, and digital healthcare. The convergence of American biotechnology and healthcare with Korean semiconductor technology is expected to yield significant synergies, highlighting the critical need for international cooperation. Joint development projects include: 1) biosensors based on biotechnology, 2) biosignal processing semiconductors based on semiconductor circuit design technology 3) artificial intelligence algorithms for effective and accurate biosignal processing.
Key Words	1. Accurate biosensors that can monitor variety of vital signs 2. Multi modal bio signal processing chip that can cover variety of biosensors 3. Low power, high resolution bio sensing solution with on device AI algorithm

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (3)	End of the Project	Level (7)
R&D Objectives	**Biosensor Technology** • Development of multiple biosensors, including non-invasive glucose sensors, PPG sensors, ECG sensors, and body impedance sensors **Chip Technology** • Development of a multimodal bio-signal processing chip capable of supporting multiple biosensors on a single die [5] • Achievement of ultra-low power consumption, small physical dimensions, high current resolution (<1 pA), and strong interference resistance **On-Device AI Embedded Chip Solution** • Integration of the bio-signal processing chip with an external AI processing component • Development of a multi-chip package (MCP) enabling on-device AI inference			
R&D Target Specifications	• Single-die chip supporting bio-impedance, ECG, PPG, and chemical sensors • Multimodal signal processing with compact form factor (≤ 3 mm × 3 mm package) • Embedded impedance boosting and noise filtering functions • Current resolution < 1 pA • Operating voltage: 1.5 V with internal DC stabilization • Internal RR detection for blood pressure estimation and advanced PPG anti-interference processing • Mixed-signal fabrication process • On-device AI support			
R&D Plan	**Phase 1: Biosensor Development** • Development of non-invasive glucose sensors, PPG sensors, ECG sensors, and body impedance sensors • Biosensor development conducted by overseas research partners • Non-invasive glucose sensing identified as a key technology **Phase 2: Multimodal Bio-Signal Processing Chip Development** • Development of a low-power, high-resolution multimodal bio-signal processing chip • Design of ultra-low-power analog and mixed-signal circuits • Support for ECG, PPG, body impedance, and chemical sensor inputs **Phase 3: On-Device AI and MCP Development** • Development of on-device AI algorithms for noise reduction, feature extraction, and early decision-making • Reduction of unnecessary data transmission to improve energy efficiency • Integration of the bio-signal processing chip with an AI processing component through MCP architecture **Phase 4: Total System Integration**			

	• Integration of biosensors, bio-signal processing chip, and on-device AI module • Component-level validation followed by system-level optimization • Optimization of signal flow, power management, and real-time processing **Development Roadmap** • **Initial Phase (TRL 3–4):** Verification of biosensor performance and core circuit technologies • **Intermediate Phase (TRL 5–6):** Prototype development with integrated multimodal sensing and on-device AI • **Final Phase (TRL 7):** System integration, reliability testing, and validation with potential customers

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	• Development of advanced biosensors, including non-invasive glucose sensors, ECG sensors, PPG sensors, body impedance sensors, and chemical sensors • Validation of biosensor accuracy and performance
Expected R&R of Korean SME Partner	• Design and fabrication of multimodal bio-signal processing chips • Development of on-device AI algorithms • MCP development and total sensing system integration

5. Expected Outcomes/Impact

Expected Outcomes and Impact	**Expected Outcomes** • Integrated biosensor, bio-signal processing chip, and on-device AI solution • Miniaturized, low-power, high-accuracy wearable biosensing platform **Market Context and Needs** The digital healthcare market is expanding rapidly [6] [7], driven by increasing demand for continuous health monitoring. However, device manufacturers face persistent challenges related to inaccurate data and limited battery life. Although AI offers significant potential to address these challenges, biological signals are inherently analog and variable, requiring specialized biosensor and signal processing technologies to bridge the gap between biological and digital domains to acquire accurate data. Accurate data acquisition is important. Just as a high-quality audio system cannot deliver great sound without a good microphone, even the most advanced cloud-based AI engine cannot produce meaningful outcomes if accurate data are not captured at the data acquisition stage. **Expected Impact**

	The proposed system combines accurate biosensors, high-resolution bio-signal processing, and on-device AI to enable intelligent decision-making at the data acquisition stage. By eliminating unnecessary data transmission and enhancing analog signal performance, the platform supports longer battery life and improved sensing reliability. The resulting technology is expected to contribute to the development of **next-generation smart wearable healthcare devices**, strengthening industrial competitiveness and advancing digital healthcare innovation.

6. References

[1] “A 114-aFrms-Resolution 46-nF/10-MΩ -Range Digital-Intensive Reconfigurable RC-to-Digital Converter with Parasitic-Insensitive Femto-Farad Baseline Sensing” Arup K.Geroge et. al. June 2018. Symposium on VLSI Circuit

[2] “A Monolithically Integrated Pressure/Oxygen/Temperature Sensing SoC for Multimodality Intracranial Neuromonitoring,” IEEE J. Solid-State Circuits (JSSC), Nov. 2014.

[3] “A 60Mb/s Wideband BCC Transceiver with 150pJ/b RX and 31pJ/b TX for Emerging Wearable Applications,” IEEE Int. Solid-State Circuits Conf. (ISSCC), Feb. 2014

[4] "A Time-domain Multi-channel Resistive-sensor Interface IC with High Energy Efficiency and Wide Input Range", *IEEE Transactions on Biomedical Circuits and Systems (**TBCAS**)*, early access, Jan. 8, 2025.

[5] “An 8MHz 31.25kS/s Impedance-Monitoring IC Based on IF Sampling Architecture with a Band Pass Delta Sigma ADC” Soon Jae Kwon et. al. June 2021, Symposium on VLSI Circuit

[6] “Global Digital Health Market Forecast” Market US, www.market.us, August 2025

[7] “Global Wearable Medical Device Market Forecast”, Moko Smart, www.mokosmart.com, Sep. 8, 2025